

# Isaac Frett

SOFTWARE ENGINEER • REMOTE ONLY, US • US CITIZEN

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## I build systems that try to prove themselves wrong.

Self-taught engineer and the primary technical owner on most of what I've built — the Python services and ETL behind executive reporting at a Fortune 50 retailer, a financial-ML platform designed to reject its own results, and a solo iOS studio taking three applications to submission. I'd rather ship one thing that survives being checked than three that look good until someone looks.

### EXPERIENCE

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#### Founder & Lead Software Engineer Jan 2026 – Present IF Development • Remote

- Founded and operate a solo iOS studio, owning product direction, architecture, implementation and the release process end to end.
- Built **three multi-platform iOS applications** for iPhone and iPad — designing the UI layouts, front-end graphics and project architecture for each. App Store submission in progress.
- Integrated push notifications, in-app purchases and AdMob SDKs through custom **Godot C++ modules** and community plugins.
- Set up the release pipeline end to end — certificates, provisioning profiles, code signing, build automation and App Store Connect submission — and ship each product with a public site that holds itself to the app's own privacy claim: no analytics, no third-party requests, so the policy stays a checkable fact rather than a promise.

Stack Godot • C++ • GDScript • iOS / iPadOS • StoreKit / IAP • AdMob • App Store Connect

#### Software Engineer Jan 2023 – Dec 2025 Lowe's Home Improvement • Charlotte, NC

- Designed and implemented scalable Python APIs (Flask, FastAPI) and microservices supporting **500+ internal employees**, with emphasis on C-level executive reporting.
- Owned the architecture and maintenance of ETL pipelines processing **~150GB of data monthly**, integrating databases into analytical models for React executive dashboards that informed **\$15M+ in business decisions**.
- Developed automations around security software for threat detection, asset patching, firewall protocols and compliance reporting.

Stack Python • Flask • FastAPI • React • SQL • ETL • microservices

#### Merchandising Service Associate Apr 2022 – Jan 2023 Lowe's Home Improvement • Coralville, IA

- Coordinated teams constructing physical marketing displays for branded products that directly impacted store revenue.
- 3D-printed a pricing sign-adaptor outside normal working hours, letting existing electronic signage be reused across any product display.

Quantitative Futures Trading Platform Archived

Sole architect and engineer · overview: github.com/isaacfrett/quant-public

End-to-end research and execution platform that discovered, validated and deployed systematic futures strategies under anti-overfitting controls, then ran them against a broker API with hard risk governance. Built so that most candidates fail.

- **Modeling.** LightGBM over a **700+ feature panel** from 50+ modular builders, on liquid CME macro futures, intraday on the 15-minute bar with 1h/4h/daily context. Two-stage side/meta architecture — a side model predicts direction, a meta model decides whether to act — with probabilities isotonically calibrated before position sizing.
- **Validation.** Walk-forward CV with **purge and embargo**, plus combinatorial purged CV for distribution-level out-of-sample estimates. Triple-barrier and volatility-normalised labels, de-overlapped so no training label reads prices inside a test window.
- **Gating.** L1 signal gates then L2 economic gates — net-of-cost Sharpe, **deflated Sharpe**, and **probability of backtest overfitting**. Commissions and slippage netted before any statistic is computed, so the gates judged tradable edge rather than paper edge.
- **Integrity as a release gate.** An automated lookahead auditor cross-checked every feature's first-valid index against the label horizon, and continuous backtest-to-live parity tests meant divergence stopped a change from shipping. Leakage was a release-blocking defect.
- **Execution and operations.** Bracketed order lifecycle with a risk governor — per-trade caps, exposure limits, drawdown halts, file-based kill switch — operated by a fleet of specialised agents across role-separated servers, surfaced through a real-time React/TypeScript dashboard over FastAPI.

Stack Python 3.11 · LightGBM · scikit-learn · pandas · NumPy · Parquet/PyArrow · FastAPI · React + TypeScript · systemd · pytest (TDD-first)

Statlume — multi-sport analytics service 2024

Solo · github.com/isaacfrett/statlume-api

- NBA, NFL and MLB statistics service over a **MySQL/SQLAlchemy warehouse** of player game logs, fed by third-party stats and sportsbook-odds APIs and run as a scheduled weekly job.
- **NBA player-prop models** for points and rebounds using TensorFlow/Keras and scikit-learn — RandomForest classifiers, standardised feature scaling, train/test splits — over rolling recent-form windows.
- **NFL player clustering** with K-means over standardised performance features for archetype segmentation, plus automated Excel datasheet generation for matchup-mismatch and three-point analysis.

Stack Python · Django · TensorFlow/Keras · scikit-learn · MySQL · SQLAlchemy · Poetry · pytest

EDUCATION

B.S. Software Development 2022 — 2024

Bellevue University · graduated, 4.0 GPA

Cybersecurity Club · Phi Theta Kappa Honors Society

SKILLS

|              |                                                                                                                                                                                                                                            |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LANGUAGES    | Python · JavaScript · TypeScript · Java · SQL · C++ · GDScript · Bash                                                                                                                                                                      |
| BACKEND      | FastAPI · Flask · Django · microservices · REST APIs · ETL pipelines                                                                                                                                                                       |
| FRONTEND     | React · TypeScript · Vite                                                                                                                                                                                                                  |
| ML LIBRARIES | LightGBM · scikit-learn · TensorFlow/Keras · PyTorch · NumPy · SciPy · pandas · PyArrow/Parquet                                                                                                                                            |
| ML METHODS   | Walk-forward CV with purge/embargo · combinatorial purged CV · meta-labeling · triple-barrier labeling · isotonic calibration · deflated Sharpe · probability of backtest overfitting · feature-leakage auditing · cost-aware model gating |
| DATA & CLOUD | PostgreSQL · MySQL · AWS · Azure · GCP                                                                                                                                                                                                     |
| DEVOPS       | Git · CI/CD (Jenkins) · Docker/Podman · Kubernetes · systemd · Grafana · Power BI · pytest                                                                                                                                                 |

Full case studies, screenshots and source links at isaacfrett.com — every claim there links to the artefact that settles it.